

Q) Discuss Factors Affecting The Time Complexity?

Ans: Time complexity refers to amount of time of an algorithm takes to execute as a function of the input size n . It depends on several factors:

1. **Input Size:** As the size of input increases, the time required by the algorithm may also increase.
2. **Nature of Input:** Different types of input may cause an algorithm to perform a different number of operations.
3. **Number of Operations:** More comparisons, assignments, arithmetic operations, etc., generally increase execution time.
4. **Loops and Iterations:** A single loop may take $O(n)$, while nested loops may take $O(n^2)$, depending on their limits.
5. **Recursive Calls:** Recursive calls can significantly affect complexity, especially when each call generates multiple additional calls.
6. **Data Structures:** The choice of data structure affects the time required for operations such as searching, insertion, deletion, and sorting.
7. **Algorithm Design:** Different algorithms solving the same problem can have different time complexities. For example, **linear search is $O(n)$** , while **binary search is $O(\log n)$** on sorted data
8. **Best, Average, and Worst Cases:** An algorithm's execution time can vary depending on the input. Therefore, complexity is often analyzed for best, average, and worst cases.

Q) Compare between Priori analysis and Posteriori analysis?

Ans:

Priori Analysis	Posteriori Analysis
1. It is done before implementing the algorithm.	1. It is done after implementing the algorithm.
2. It is a theoretical analysis.	2. It is an experimental/practical analysis.
3. It calculates time and space complexity mathematically.	3. It measures the actual execution time and memory used.
4. It is independent of the computer and programming language.	4. It depends on the computer, programming language, compiler, and system.
5. It is useful for comparing algorithms theoretically.	5. It is useful for evaluating actual performance.
6. It does not require the algorithm to be implemented.	6. It requires the algorithm to be implemented and tested.
7. It uses measures such as Big-O, Big- Ω and Big- Θ .	7. It uses actual measurements obtained from program execution.
8. It can be done without writing the complete program.	8. Its results are machine-dependent.